Technological Knowledge Intensity and Intergenerational Mobility across U.S. Regions

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May 31, 2026

Abstract

This paper studies the regional relationship between technological knowledge intensity and intergenerational upward mobility in the United States. Combining mobility measures from the Opportunity Atlas with U.S. patent records, I measure upward mobility as the expected adult income rank of children who grew up in a commuting zone with parents at the 25th percentile of the national income distribution. Technological knowledge intensity is measured using patent citations per capita at the commuting-zone level. Across U.S. commuting zones, regions with higher patent-based knowledge intensity exhibit higher upward mobility. A historical knowledge-spillover exposure measure, constructed from pre-existing patent citation links across regions, yields similar positive estimates. The evidence is consistent with the view that technologically dynamic regions are associated with broader opportunity structures.

JEL Codes: J62, O31, O33, R11, R23
Keywords: technological knowledge, patent citations, intergenerational mobility, commuting zones, knowledge spillovers, regional innovation

1 Introduction

Technological change is a central source of long-run productivity growth and rising living standards. Yet its social consequences are uneven. A large literature studies how innovation affects productivity, firm dynamics, wages, and inequality (Schumpeter, 1942; Solow, 1956; Romer, 1990; Kogan et al., 2017; Aghion et al., 2019). Much less is known about whether technologically dynamic regions are also associated with broader economic opportunity across generations. This paper studies

*KU Leuven, sebastien.lamproye@kuleuven.be. Acknowledgements: I would like to thank Philippe Aghion, Anders Åkerman, Lorenzo Cassi, Mridul Joshi, Miklós Koren, Patrick Legros, David Margolis, Jaime Mauricio Montana Doncel, Sumit Parashar, Duy Anh Dang Phan, Juliana Pinillos, Thiago Scarelli, Angelo Secchi and the participants of the seminars at the Central European University - Department of Economics and Business, Collège de France, ECARES-ULB, INSEAD, Paris School of Economics, Stockholm School of Economics, Université Catholique de Louvain, Université de Namur and Université Saint-Louis for their support and helpful suggestions. Finally, I gratefully acknowledge financial support from FWO (grant 11PD024N).

that question by examining the relationship between regional technological knowledge intensity and intergenerational upward mobility in the United States.

The outcome of interest is intergenerational income rank mobility. I use the absolute upward mobility measure from the Opportunity Atlas, commonly referred to as AM25 (Chetty et al., 2014). For each commuting zone, AM25 is the expected adult income rank of children who grew up in that location with parents at the 25th percentile of the national income distribution. This measure is not a transition probability and does not track movement into a particular occupation or job ladder. Instead, it summarizes the long-run adult income outcomes of children from relatively low-income families, assigning them to the commuting zone in which they grew up. It is therefore well suited for studying whether some local environments are associated with better economic prospects for children from disadvantaged backgrounds.

The analysis focuses on commuting zones because they approximate local labor markets and because both innovation and economic opportunity vary sharply across space. Innovative activity is geographically concentrated and embedded in local networks of firms, universities, skilled workers, suppliers, and knowledge flows (Jaffe et al., 1993; Jaffe and Trajtenberg, 2002; Audretsch and Feldman, 1996; Anselin et al., 1997; Breschi and Lissoni, 2001; Feldman and Kogler, 2010; Carlino and Kerr, 2015). Intergenerational mobility also differs substantially across places in the United States (Chetty et al., 2014, 2016; Hendren, 2018; Connolly et al., 2019; Chetty et al., 2020). These two facts motivate the central question of the paper: are regions with greater technological knowledge intensity also regions where children from low-income families achieve higher adult income ranks?

To measure technological knowledge intensity, I use patent citations per capita at the commuting-zone level. Patent citations are widely used in the economics of innovation because they contain information about the technological relevance of patented inventions and the diffusion of knowledge through the patent system (Trajtenberg, 1990; Hall et al., 2005; Jaffe and Trajtenberg, 2002). At the same time, they are an imperfect measure of innovation. Patent citations do not capture all forms of innovation, commercialization, organizational change, or locally implemented process improvements. They are also affected by differences in patenting propensities, technology fields, citation practices, and the distinction between private and social value (Griliches, 1990; Lanjouw and Schankerman, 2004; van Zeebroeck, 2011). For this reason, I interpret the main explanatory variable as patent-based technological knowledge intensity rather than as a complete measure of innovation in all its forms.

The empirical analysis is cross-sectional at the commuting-zone level. I combine mobility outcomes from the Opportunity Atlas with patent records aggregated by inventor location. The baseline specification relates AM25 to patent-citation intensity, controlling for commuting-zone characteristics such as demographic composition, education, industrial structure, local economic conditions, and migration flows. These estimates document whether technologically intensive regions are also regions with higher upward mobility for children from low-income families.

A central empirical challenge is that the relationship between technological knowledge intensity and mobility may not be causal. Regions with higher mobility may attract innovative firms and workers. Conversely, regions with strong universities, high human capital, favorable institutions, or historically dynamic industries may produce both more patent citations and higher upward mobility. To provide suggestive evidence beyond cross-sectional correlations, I use a knowledge-spillover exposure measure based on historical patent citation links across commuting zones. The instrument predicts local technological knowledge intensity using innovation in other regions, weighted by the extent to which inventors in the focal commuting zone historically cited patents from those external regions.

This strategy builds on the idea that patent citation networks reveal channels of knowledge diffusion across places (Jaffe et al., 1993; Jaffe and Trajtenberg, 2002; Aghion et al., 2019). The historical citation network is measured using patent citations from 1993–1995, before the main innovation measure. Regions that historically drew more heavily on ideas from particular external locations are therefore more exposed to subsequent technological developments in those locations. The direction of the instrument is important: the relevant weights capture the propensity of inventors in a focal commuting zone to cite patents from other commuting zones, so later innovation in those historically cited external regions predicts technological knowledge intensity in the focal region.

The main finding is that commuting zones with higher patent-based technological knowledge intensity tend to have higher intergenerational upward mobility. A one-standard-deviation increase in technological knowledge intensity is associated with approximately a 0.06-standard-deviation increase in AM25. Given the standard deviation of AM25 in the sample, this corresponds to roughly one third of an income-rank point. The magnitude is modest, but it suggests that technological dynamism is positively related to the geography of economic opportunity. The spillover-based IV estimates are also positive, consistent with the interpretation that regional exposure to external knowledge flows is associated with higher upward mobility through local technological dynamism.

The paper contributes to three related literatures. First, it contributes to work on intergenerational mobility and place by highlighting the potential role of local productive structure and technological dynamism as features of opportunity environments (Chetty et al., 2014, 2016; Hendren, 2018; Connolly et al., 2019). Second, it contributes to the literature on the geography of innovation and knowledge spillovers by connecting patent-based measures of regional technological activity to a long-run social outcome beyond productivity, wages, or firm performance (Jaffe et al., 1993; Audretsch and Feldman, 1996; Breschi and Lissoni, 2001; Feldman and Kogler, 2010; Carlino and Kerr, 2015). Third, it relates to work on innovation and inequality by shifting attention from income dispersion and top income shares to intergenerational upward mobility (Piketty, 2014; Jones and Kim, 2018; Frydman and Papanikolaou, 2018; Aghion et al., 2019).

The paper is closest to Aghion et al. (2019), who study innovation and top income inequality and also report evidence relating innovation to social mobility. The present paper differs by placing intergenerational upward mobility at the center of the analysis, using commuting-zone AM25 as

the main outcome, and applying a historical citation-network exposure design to study whether patent-based technological knowledge intensity is associated with broader regional opportunity. In this sense, the contribution is not only to revisit the innovation–mobility correlation, but to connect the geography of technological knowledge to the geography of upward mobility using predetermined knowledge-link exposure across regions.

The interpretation of the results is intentionally cautious. One possible mechanism is that technologically dynamic regions generate broader opportunity structures, including applied technical tasks, skilled production work, and other forms of employment that may provide upward career paths for individuals without elite credentials. This interpretation is consistent with research on technological change, tasks, and labor demand (Autor et al., 2003; Acemoglu and Autor, 2011; Beaudry et al., 2016). However, the paper does not observe individual occupational trajectories or directly measure changes in middle-skill employment. The evidence should therefore be read as documenting a regional relationship between technological knowledge intensity and upward mobility, with the labor-market mechanism remaining suggestive.

The rest of the paper proceeds as follows. Section 2 presents the conceptual motivation. Section 3 describes the data and measurement. Section 4 presents the empirical strategy. Section 5 reports and interprets the empirical results. Section 6 concludes.

2 Conceptual Motivation: Technological Dynamism and Opportunity Structures

This section provides a simple conceptual motivation for why regional technological knowledge intensity may be associated with intergenerational upward mobility. The purpose is not to develop a structural model or to establish a mechanism formally. Rather, the goal is to clarify how technologically dynamic local economies could be related to the adult income ranks of children from low-income families.

The starting point is that commuting zones differ in their local opportunity structures. Some regions offer mainly low-wage or low-progression jobs, while others contain denser networks of firms, suppliers, universities, and technical occupations. Research on the geography of innovation emphasizes that technological activity is spatially concentrated and shaped by localized knowledge spillovers, agglomeration economies, and regional innovation systems (Jaffe et al., 1993; Audretsch and Feldman, 1996; Anselin et al., 1997; Breschi and Lissoni, 2001; Cooke et al., 1997; Carlino and Kerr, 2015). These regional differences may matter for upward mobility because children's long-run outcomes are strongly related to where they grow up (Chetty et al., 2014; Hendren, 2018).

Technologically dynamic regions may generate a broader set of employment opportunities through several channels. New products and processes can increase demand for implementation, maintenance, logistics, quality control, production support, and applied problem-solving. These tasks may

create or expand jobs that do not necessarily require elite credentials but can still offer higher earnings trajectories than routine low-wage work. This interpretation is consistent with work showing that technological change reshapes task demand, occupational structure, and the returns to different types of skills (Autor et al., 2003; Acemoglu and Autor, 2011; Beaudry et al., 2016). It is also consistent with broader evidence that local labor-market shocks and local job structures can have persistent effects on regional economic outcomes (Moretti, 2012; Autor et al., 2013).

This idea connects technological dynamism to mobility through access to opportunity rather than through redistribution. Intergenerational upward mobility, as measured by AM25, captures the expected adult income rank of children from families at the 25th percentile of the national income distribution. A region may therefore display higher upward mobility if children from low-income families face a higher probability of reaching occupations, firms, or career paths associated with higher adult income ranks. In this interpretation, technological knowledge intensity matters because it may shape the set of attainable local opportunities.

The distinction between mobility and inequality is important. Innovation may increase income dispersion if it generates large returns for inventors, entrepreneurs, executives, or highly skilled workers (Jones and Kim, 2018; Frydman and Papanikolaou, 2018; Aghion et al., 2019). At the same time, it may also expand applied technical, skilled production, or support roles that improve career prospects for individuals from less advantaged backgrounds. Thus, innovation and technological dynamism could be associated with both higher inequality and higher upward mobility. Whether this occurs depends on the types of jobs created, the accessibility of those jobs, and the local institutions that shape education, training, matching, and labor-market entry.

The empirical analysis does not directly observe these mechanisms. It does not follow individuals across occupations, measure transitions into technical jobs, or estimate how local occupational structures change in response to innovation. The mechanism should therefore be interpreted as suggestive. The main empirical question is more modest: whether commuting zones with greater patent-based technological knowledge intensity also exhibit higher intergenerational upward mobility.

This interpretation also clarifies the role of the knowledge-spillover instrument. Historical citation links across regions are used to construct exposure to external technological developments. If these spillovers raise local patent-based technological knowledge intensity, they may also be associated with changes in local opportunity structures. However, the exclusion restriction remains demanding, because historical knowledge networks may also proxy for persistent regional characteristics such as universities, human capital, industrial specialization, or long-run growth capacity. The empirical estimates should therefore be read as suggestive evidence on the relationship between technological dynamism and mobility, not as definitive proof of a specific job-ladder mechanism.

The conceptual implication is that regional technological dynamism may matter for intergenerational mobility when it broadens the set of accessible career paths for children from low-income

families. This channel is likely to be strongest in places where technological activity is connected to local labor demand and where education, training, housing, and labor-market institutions allow disadvantaged children to access emerging opportunities. Conversely, technological dynamism may have weaker mobility effects if its benefits accrue mainly to highly educated workers, incumbent elites, or in-migrants from other regions.

3 Data and Measurement

This section describes the data sources, variable construction, and sample used in the empirical analysis. The unit of observation is the commuting zone (CZ), which approximates local labor markets and is commonly used to study spatial variation in economic opportunity. The analysis combines commuting-zone measures of intergenerational mobility with patent-based measures of regional technological knowledge intensity.

3.1 Intergenerational mobility outcomes

Measures of intergenerational mobility are drawn from the Opportunity Atlas (Chetty et al., 2014). The main outcome variable is absolute upward mobility, commonly referred to as AM25. For each commuting zone, AM25 is defined as the expected national income rank in adulthood of children who grew up in that commuting zone with parents at the 25th percentile of the national income distribution.

This measure has three features that are important for interpretation. First, AM25 is an income-rank outcome, not a measure of income levels or local wage inequality. Second, it is assigned according to childhood location, so it captures the long-run adult outcomes of children who grew up in a given commuting zone. Third, it is an aggregate place-level measure: the data do not identify individual occupational trajectories, the specific firms employing individuals, or whether individuals remain in their childhood commuting zone as adults. The analysis therefore studies regional differences in upward mobility rather than individual-level transitions into particular jobs or occupations.

AM25 is well suited to the question studied here because it focuses on children from relatively low-income families. A higher value of AM25 means that children who grew up in a commuting zone with parents at the 25th percentile reach a higher expected rank in the national adult income distribution. This makes the measure useful for studying whether technologically dynamic regional economies are associated with better long-run outcomes for children from disadvantaged backgrounds.

3.2 Technological knowledge intensity

Technological knowledge intensity is measured using patent data from the U.S. Patent and Trademark Office. The main measure is patent citations received within five years of grant, aggregated to the commuting-zone level based on inventor location and scaled by population. I average this citation-per-capita measure over patents granted between 2001 and 2010.

Patent citations are widely used in the economics of innovation because they provide information on the technological relevance of patented inventions and on the diffusion of knowledge through the patent system (Trajtenberg, 1990; Hall et al., 2005; Jaffe and Trajtenberg, 2002). Using citations rather than simple patent counts gives greater weight to patents that are more heavily used by later inventions.

At the same time, citation-based measures are imperfect indicators of innovation. Patent citations do not capture all economically relevant innovation, including non-patented process improvements, organizational changes, informal know-how, and local adoption of existing technologies. Patenting propensities also vary across sectors and regions, and citation practices differ across technology classes (Griliches, 1990). Moreover, patent quality is multidimensional, and alternative indicators such as claims, renewals, and patent family size may capture different dimensions of private value or commercialization (Lanjouw and Schankerman, 2004). Evidence that patent-value indicators do not always rank inventions similarly further supports a cautious interpretation of citation-based measures (van Zeebroeck, 2011).

For these reasons, I interpret the measure as patent-based technological knowledge intensity rather than as a complete measure of innovation or commercialization. The empirical analysis therefore asks whether regions with greater patent-citation intensity are also regions with higher upward mobility.

For patents with multiple inventors located in different commuting zones, citations are fractionally allocated across inventor locations. This avoids mechanically assigning the full value of a patent to a single region and better reflects the geographic distribution of inventive activity.

The timing of the measure should also be interpreted cautiously. The patent-citation measure is averaged over 2001–2010 and is intended to capture sustained regional technological dynamism rather than a sharply timed innovation shock. The mobility outcome reflects long-run adult outcomes of cohorts assigned to childhood commuting zones. The empirical design is therefore best interpreted as comparing regions with different levels of technological knowledge intensity and upward mobility.

3.3 Knowledge-spillover exposure

To provide evidence beyond ordinary cross-sectional correlations, I construct a knowledge-spillover exposure measure based on historical patent citation links across commuting zones. The measure uses citations observed between 1993 and 1995 to capture pre-existing knowledge linkages across

regions.

For each commuting zone, I compute a weighted sum of technological knowledge intensity in other commuting zones, where the weights reflect the historical propensity of inventors in the focal commuting zone to cite patents from those external regions. Thus, the direction of the measure is from the focal commuting zone to the external regions it historically cited. If inventors in commuting zone c historically drew heavily on patents from commuting zone c', then later technological activity in c' receives a higher weight in the spillover exposure measure for c.

Formally, the spillover exposure measure for commuting zone c is

$$\text{SpilloverIV}_c = \sum_{c' \neq c} w_{cc'} \, K_{c'}, \tag{1}$$

where $w_{cc'}$ is the share of historical patent citations from commuting zone c to commuting zone c' in the 1993–1995 citation network, and $K_{c'}$ denotes subsequent patent-based technological knowledge intensity in commuting zone c'. Local technological knowledge intensity is excluded from the construction of the exposure measure.

The purpose of this variable is to capture exposure to external technological developments through historically observed knowledge links. Because the citation weights are measured before the main patent-citation period, the measure reduces concerns about contemporaneous reverse causality.

3.4 Controls, weights, and sample

The regressions include commuting-zone controls capturing demographic composition, educational attainment, industrial structure, and local economic conditions. These controls account for factors that may be correlated with both technological knowledge intensity and intergenerational mobility, such as human capital, labor-market composition, urbanization, and baseline economic conditions. Some specifications additionally include migration controls to account for differences in population turnover and selective sorting across regions.

All regressions are weighted by commuting-zone population in 2000. Standard errors are clustered at the state level to allow for spatial correlation in unobserved shocks across commuting zones within the same state. Unless otherwise stated, continuous variables are standardized to have mean zero and unit variance, which allows the estimated coefficients to be interpreted in standard-deviation units.

Table 1 reports summary statistics for the main variables used in the analysis. The final sample consists of 669 commuting zones for which the mobility outcome, patent-based technological knowledge measure, spillover exposure measure, and control variables are available.

Table 1: Summary Statistics

	Mean	Std. Dev.	Min	Max
AM25 (income rank)	43.98	5.59	33.14	64.02
Technological knowledge intensity (citations per capita)	0.00078	0.00137	0.00000	0.02321
Test Scores (income-adjusted)	0.26	7.75	-31.84	20.07
Social Capital Index	0.09	1.22	-3.20	5.27
Manufacturing employment share	0.15	0.08	0.01	0.44

Notes: This table reports summary statistics at the commuting-zone (CZ) level. AM25 is the expected income rank of children whose parents are at the 25th percentile of the national income distribution. Technological knowledge intensity is measured as patent citations received within five years of grant per capita, averaged over 2001–2010.

4 Empirical Strategy

This section describes the empirical approach used to study the relationship between regional technological knowledge intensity and intergenerational upward mobility. The analysis proceeds in three steps. First, I estimate cross-sectional ordinary least squares specifications that document the association between patent-based technological knowledge intensity and AM25 across commuting zones. Second, I use a knowledge-spillover exposure measure as an instrumental variable based on predetermined citation links across regions. Third, I discuss the identifying interpretation and the scope of the estimates.

4.1 Baseline specification

I begin by estimating the following cross-sectional specification at the commuting-zone level:

$$\text{AM25}_c = \alpha + \beta K_c + \mathbf{X}_c'\gamma + \varepsilon_c, \tag{2}$$

where AM25c denotes absolute upward mobility in commuting zone c, measured as the expected national income rank in adulthood of children whose parents were at the 25th percentile of the national income distribution. The variable K_c denotes patent-based technological knowledge intensity, measured by patent citations received within five years per capita and averaged over patents granted between 2001 and 2010. The vector $\mathbf{X}_c$ includes commuting-zone controls capturing demographic composition, educational attainment, industrial structure, local economic conditions, and, in some specifications, migration flows.

The coefficient β captures the conditional association between technological knowledge intensity and upward mobility across commuting zones. Because both AM25c and Kc are standardized in the main specifications, β can be interpreted in standard-deviation units. This baseline specification documents whether regions with greater patent-based technological knowledge intensity also exhibit higher upward mobility for children from low-income families.

OLS estimates may reflect several forces. Regions with higher mobility may attract innovative firms, skilled workers, or research institutions. Conversely, persistent regional characteristics such as universities, industrial specialization, human capital, institutions, or long-run growth trajectories may jointly shape patenting activity and mobility outcomes. These concerns motivate the spillover-based instrumental-variable specification below.

4.2 Spillover-based instrumental-variable specification

To move beyond ordinary cross-sectional correlations, I use the knowledge-spillover exposure measure described in Section 3.3 as an instrument for local technological knowledge intensity. The instrument uses historical patent citation links to predict technological knowledge intensity in a focal commuting zone based on subsequent technological activity in other commuting zones to which the focal zone was historically connected.

The two-stage least squares specification is

$$\begin{aligned} K_c &= \pi \text{SpilloverIV}_c + \mathbf{X}_c'\delta + u_c, \\ \text{AM25}_c &= \alpha + \beta \widehat{K}_c + \mathbf{X}_c'\gamma + \varepsilon_c. \end{aligned} \tag{3}$$

where SpilloverIV_c is the exposure to external technological knowledge flows constructed from the 1993–1995 citation network, and $\widehat{K}_c$ is the fitted value of technological knowledge intensity from the first stage.

The first stage captures whether commuting zones historically connected to external sources of technological knowledge subsequently exhibit higher local patent-based knowledge intensity. The second stage relates upward mobility to the component of local technological knowledge intensity predicted by this historical spillover exposure.

4.3 Identification and interpretation

The spillover-based IV strategy relies on two conditions. First, the instrument must be relevant: historical citation links to external regions must predict subsequent patent-based technological knowledge intensity in the focal commuting zone. This condition can be assessed empirically through the first-stage estimates and first-stage F-statistics reported in Table 3.

Second, the identifying interpretation requires that historical exposure to external technological developments affects intergenerational upward mobility primarily through subsequent local technological knowledge intensity. The design strengthens the comparison relative to ordinary cross-sectional correlations because the citation links are predetermined, local technological knowledge intensity is excluded from the exposure measure, and the regressions control for a range of commuting-zone characteristics.

At the same time, historical citation networks may partly reflect persistent regional characteristics, such as research universities, sectoral specialization, human capital, firm networks, or long-run industrial trajectories. The estimates should therefore be interpreted as spillover-based evidence on the relationship between technological knowledge intensity and upward mobility, rather than as estimates that identify all possible channels through which technological dynamism affects opportunity.

This interpretation also clarifies the scope of the estimates. The IV variation captures differences in technological knowledge intensity predicted by historical citation-based exposure to other regions. It does not capture all sources of local technological activity, nor does it identify a specific individual-level mechanism. In particular, the analysis cannot determine whether the children who experience upward mobility are the same individuals who directly benefit from local technological activity, nor can it distinguish between channels operating through occupational structure, education, migration, local prices, or other regional characteristics.

4.4 Inference

All regressions are weighted by commuting-zone population in 2000 to account for differences in the size of local labor markets. Standard errors are clustered at the state level to allow for spatial correlation in unobserved shocks across commuting zones within states. Unless otherwise stated, continuous variables are standardized to have mean zero and unit variance, so estimated coefficients can be interpreted in standard-deviation units.

Throughout the analysis, I report first-stage statistics to assess instrument relevance. The statistical significance and magnitude of the estimates should be interpreted alongside the regional level of analysis and the identifying interpretation described above.

5 Results and Interpretation

This section presents the empirical relationship between patent-based technological knowledge intensity and intergenerational upward mobility across U.S. commuting zones. I first report the baseline cross-sectional associations and then turn to the spillover-based IV estimates. I then discuss the magnitude of the estimates and the main limitations of interpretation.

Table 2: Technological Knowledge Intensity and Intergenerational Mobility

	(1) OLS	(2) IV + Controls	(3) IV + Controls + Migration
Technological knowledge intensity	0.059***	0.045*	0.057**
	(0.016)	(0.025)	(0.027)
Controls	No	Yes	Yes
Migration controls	No	No	Yes
Observations	669	669	669
R^2	0.034	0.645	0.651

Notes: The dependent variable is AM25. Technological knowledge intensity is measured as citations received within five years of grant per capita, averaged over 2001–2010. All variables except the urban indicator are standardized. Regressions are weighted by CZ population (year 2000). Standard errors clustered by state are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.1 Baseline associations

Column (1) of Table 2 reports the unconditional relationship between technological knowledge intensity and absolute upward mobility. The estimated coefficient is positive and statistically significant, indicating that commuting zones with higher patent-citation intensity tend to exhibit higher upward mobility for children from low-income families.

Columns (2) and (3) progressively add commuting-zone controls capturing demographic characteristics, educational attainment, industrial composition, labor-market conditions, and migration flows. The coefficient remains positive after including these controls. These estimates suggest that the relationship between patent-based technological knowledge intensity and upward mobility is not driven solely by the most basic observable differences across regions.

The OLS estimates are descriptive. They do not establish that technological knowledge intensity causes higher mobility. Regions with stronger mobility may attract innovative firms and workers, and persistent regional characteristics may jointly shape both technological activity and mobility outcomes. For this reason, the next subsection reports spillover-based IV estimates that use historical knowledge links across regions as a source of suggestive variation.

5.2 Spillover-based IV estimates

Table 2 also reports two-stage least squares estimates using the knowledge-spillover exposure measure as an instrument for local technological knowledge intensity. The IV estimates remain positive. This pattern is consistent with the interpretation that regions whose technological knowledge intensity is predicted by historical exposure to external knowledge flows also exhibit higher intergenerational upward mobility.

First-stage results, reported in Table 3, show that the knowledge-spillover exposure measure is

strongly related to local patent-based technological knowledge intensity. The first-stage F-statistics exceed conventional weak-instrument thresholds, suggesting that weak-instrument concerns are unlikely to drive the IV results.

Table 3: First Stage: Knowledge-Spillover Exposure and Technological Knowledge Intensity

	(1) Reduced	(2) IV + Controls	(3) IV + Controls + Migration
Knowledge-spillover IV	0.659***	0.795***	0.792***
	(0.161)	(0.177)	(0.177)
Observations	669	669	669
First-stage F	16.8	20.2	20.0
R^2	0.555	0.762	0.773

Notes: The dependent variable is patent-based technological knowledge intensity. The instrument is the knowledge-spillover index constructed from the 1993–1995 patent citation network. All regressions are weighted by CZ population (year 2000). Standard errors clustered by state are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.3 Magnitude

The estimated magnitude is modest but economically interpretable. A one-standard-deviation increase in patent-based technological knowledge intensity is associated with approximately a 0.06-standard-deviation increase in AM25. Since the standard deviation of AM25 in the sample is about 5.6 income-rank points, this corresponds to roughly one third of an income-rank point.

This magnitude should be interpreted in light of the regional and cross-sectional nature of the analysis. The estimates do not imply that innovation policy alone would generate large changes in intergenerational mobility. Rather, they suggest that technological dynamism is positively related to the geography of opportunity and may be one component of the broader local environment shaping the long-run outcomes of children from low-income families.

5.4 Interpretation and scope

The results are consistent with the idea that technologically dynamic regions are associated with broader opportunity structures. One possible mechanism is that local technological activity increases demand for applied technical tasks, skilled production work, maintenance, logistics, quality control, and other activities that may create upward career paths outside elite occupations. The empirical analysis does not directly test this mechanism, because it does not observe individual occupational histories, transitions into technical jobs, or changes in local occupational structure.

The estimates should therefore be interpreted as regional relationships rather than individual-level effects. The mobility outcome is assigned by childhood location, while adult income may be

realized elsewhere. As a result, local technological dynamism may matter through several channels, including schooling, expectations, networks, early labor-market exposure, parental opportunities, migration, local public finances, or adult employment opportunities.

Two scope conditions are important. First, the patent-citation measure captures patent-based technological knowledge intensity, not innovation in all its forms. It does not directly measure non-patented innovation, organizational change, commercialization, or local technology adoption. Second, the spillover-based design identifies variation associated with historical citation exposure. This strengthens the comparison relative to ordinary cross-sectional correlations, while still requiring that historical citation links affect mobility primarily through subsequent local technological knowledge intensity rather than through other persistent regional characteristics.

Taken together, the evidence suggests that technological knowledge intensity is positively related to the geography of intergenerational opportunity. The findings are consistent with a labor-market opportunity interpretation, but they also leave room for alternative channels and call for future work using individual-level data, occupational histories, or panel variation to identify the mechanisms more directly.

6 Conclusion

This paper studies the regional relationship between patent-based technological knowledge intensity and intergenerational upward mobility in the United States. Combining commuting-zone mobility measures from the Opportunity Atlas with patent citation records, the analysis shows that regions with higher technological knowledge intensity tend to exhibit higher adult income ranks for children who grew up in low-income families.

The evidence is positive across both baseline cross-sectional specifications and spillover-based IV specifications. The estimated magnitude is modest: a one-standard-deviation increase in patent-based technological knowledge intensity is associated with roughly a 0.06-standard-deviation increase in AM25, or about one third of an income-rank point. These results suggest that technological dynamism is related to the geography of economic opportunity.

The interpretation should reflect the regional level of analysis. The paper studies commuting zones rather than individual workers, and it does not observe individual occupational histories, job transitions, or direct access to technology-related employment. The patent-citation measure captures technological knowledge intensity within the patent system, not all forms of innovation, commercialization, organizational change, or local technology adoption. The spillover-based design strengthens the comparison relative to ordinary cross-sectional correlations by using predetermined citation-network exposure, while the identifying interpretation still requires that these historical knowledge links operate primarily through subsequent local technological knowledge intensity rather than through other persistent regional characteristics.

One possible interpretation is that technologically dynamic regions provide broader opportunity structures, including applied technical tasks, skilled production work, and other career paths that may be accessible to individuals without elite credentials. This mechanism is consistent with the idea that local productive structure and technological dynamism can shape the opportunities available to children from disadvantaged backgrounds. However, the mechanism remains suggestive because the paper does not directly measure occupational transitions or changes in middle-skill employment.

The findings therefore contribute to the literature by linking the geography of innovation to the geography of intergenerational mobility. While prior work has emphasized the relationship between innovation, growth, and inequality, this paper highlights upward mobility as a distinct outcome. The results suggest that regional technological dynamism may be one component of the broader opportunity environment, alongside schools, neighborhoods, social capital, housing markets, and labor-market institutions.

The policy implications should also be interpreted cautiously. The results do not imply that innovation policy alone can substantially raise intergenerational mobility. Rather, they suggest that the opportunity effects of technological dynamism are likely to depend on complementary institutions that allow children from low-income families to access emerging opportunities. Policies related to education, training, housing, mobility, and labor-market access may therefore shape whether technological dynamism translates into broader economic opportunity.

Future research could strengthen the evidence by linking innovation exposure to individual-level data on education, occupations, earnings trajectories, and migration. Panel variation or policy-induced changes in regional technological activity would also help clarify the causal channels. Such work would make it possible to distinguish more directly between mechanisms operating through occupational structure, local labor demand, schooling, migration, public finances, and housing markets.

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Appendix

A Data Construction and Variable Definitions

This section provides additional detail on the construction of the main variables used in the empirical analysis.

A.1 Intergenerational mobility

Intergenerational mobility outcomes are drawn from the Opportunity Atlas. The main outcome variable is absolute upward mobility (AM25), defined as the expected national income rank in adulthood of children whose parents were at the 25th percentile of the national income distribution. Measures are computed at the commuting-zone level and assigned according to childhood location. They therefore reflect long-run adult outcomes of cohorts growing up in each commuting zone.

A.2 Technological knowledge intensity

Technological knowledge intensity is measured using patent data from the U.S. Patent and Trademark Office. The main measure is based on patent citations received within five years of grant, aggregated to the commuting-zone level using inventor location and scaled by population. The measure averages citations per capita over patents granted between 2001 and 2010. For patents with inventors in multiple commuting zones, citations are fractionally allocated across inventor locations. The resulting variable is standardized to have mean zero and unit variance in the main specifications.

A.3 Controls

Controls capture commuting-zone demographic composition, educational attainment, industrial structure, and local economic conditions. These include population size, education shares, employment composition by sector, urbanization indicators, and migration measures. All continuous control variables are standardized.

Table 4 lists the commuting-zone controls used in the main specifications.

Table 4: List of controls

Control	**Definition / source (commuting-zone level)**
Population (2000)	Total population; used for regression weights.
Urbanization indicator	Indicator for urban CZ (or share urban), based on Census/commuting-zone definition.
Education: HS share	Share of adults with at least a high-school degree.
Education: college share	Share of adults with at least a bachelor's degree.
Income level / log income	Baseline income level measure (e.g., log per-capita income).
Unemployment rate	Local unemployment rate.
Labor force participation	Share of working-age population in the labor force.
Employment shares by sector	Shares employed in broad industries (e.g., manufacturing, services, construction, public sector).
Racial composition	Shares by race/ethnicity (e.g., White, Black, Hispanic, Other).
Foreign-born share	Share of residents born outside the U.S.
Family structure	Measures such as single-parent share or marriage rate (as available).
Residential segregation	Segregation indices (if included in your baseline controls).
Social capital / civic measures	Social capital proxy measures (if included).
Migration / turnover	Measures of in- and out-migration or population turnover (if included).

Notes: The table reports the main control categories included in the baseline commuting-zone specifications. All continuous controls are standardized (mean zero, unit variance) unless otherwise stated.

B Instrument Construction: Knowledge-Spillover Exposure

This section provides additional detail on the construction of the knowledge-spillover exposure measure used in the instrumental-variable analysis.

The measure exploits historical patent citation networks to predict local patent-based technological knowledge intensity using technological activity originating in other commuting zones. Citation links across commuting zones are computed using patent citations observed between 1993 and 1995. These citation shares capture the historical propensity of inventors in a given commuting zone to cite patents originating in other zones.

For each commuting zone, the spillover exposure measure is constructed as a weighted sum of technological knowledge intensity in other commuting zones, where weights correspond to historical citation shares. Technological knowledge intensity in external zones is measured using patent citations over horizons ranging from zero to ten years. Local technological knowledge intensity is excluded by construction.

Formally, the spillover exposure measure for commuting zone c is

$$\text{SpilloverIV}_c = \sum_{c' \neq c} w_{cc'} K_{c'}, \tag{4}$$

where $w_{cc'}$ denotes the share of citations from commuting zone c to commuting zone c' in the 1993–1995 citation network, and $K_{c'}$ denotes patent-based technological knowledge intensity in external commuting zone c'.

Because the citation network predates the main patent-citation measure and because the exposure measure excludes local technological knowledge intensity, it reduces concerns about contemporaneous reverse causality. However, as discussed in the main text, historical citation patterns may also reflect persistent regional characteristics such as universities, human capital, sectoral specialization, research capacity, or long-run industrial trajectories. The instrument should therefore be interpreted as a source of suggestive variation rather than as mechanically exogenous.

C Estimation Details and Inference

All regressions are estimated at the commuting-zone level and weighted by population in 2000. Standard errors are clustered at the state level to account for spatial correlation in unobserved shocks across commuting zones within the same state. Continuous variables are standardized in the main specifications unless otherwise stated. Instrument strength is assessed using first-stage F-statistics.

D Replication Package

The analysis is replicable using the data construction and estimation files associated with the project. The replication package documents the construction of the commuting-zone mobility measures, patent-based technological knowledge measures, spillover exposure measure, controls, and regression tables.